POLITECNICO
MILANO 1863

Aerospace Control Systems
Exam Project AY 2022/23

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Summary

- Introduction
- Tasks
  - Task 1: Attitude Control
  - Task 2: Robustness Analysis
  - Task 3: Dynamics of the Observer
  - Task 4: Monte Carlo Validation

- Task 1 – Attitude Control

Introduction

- Project goal is to design and verify a robust single-axis attitude control system for ANT-X quadrotor drone.

- Lateral Dynamics

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

$$x = \begin{bmatrix} v \\ p \\ \varphi \end{bmatrix} \qquad u = \delta_{lat}, \qquad y = \begin{bmatrix} p \\ \varphi \\ a_y \end{bmatrix}$$

$$A = \begin{bmatrix} Y_v & Y_p & g \\ L_v & L_p & 0 \\ 0 & 1 & 0 \end{bmatrix}, B = \begin{bmatrix} Y_\delta \\ L_\delta \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ Y_v & Y_p & 0 \end{bmatrix}, D = \begin{bmatrix} 0 \\ 0 \\ Y_\delta \end{bmatrix}$$

Introduction

- Given Data

Stability derivatives

$Y_v = -0.1068 \ 1/s \ (4.26\%)$

$Y_p = 0.1192 \ m/s \ rad \ (2.03\%)$

$L_v = -5.9755 \ rad \ s/m \ (1.83\%)$

$L_p = -2.6478 \ 1/s \ (2.01\%)$

Control derivatives

$Y_d = -10.1647 \ m/s^2 \ (1.37\%)$

$L_d = 450.7085 \ rad/s^2 \ (0.81\%)$

- Uncertainty is given in terms of standard deviations assuming a Gaussian density for each parameter

Control-Oriented Assumptions

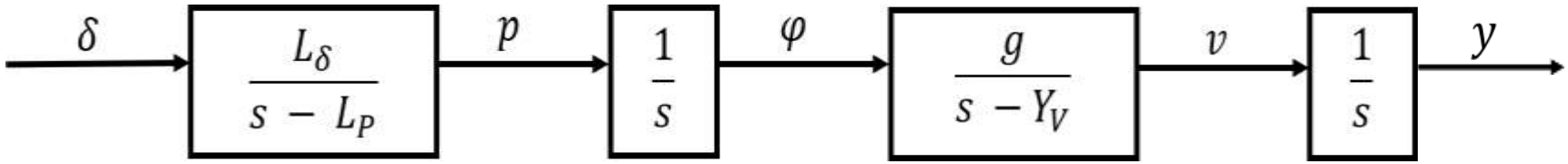
Control-Oriented Assumptions:

Y_p= 0 m/s rad

L_v= 0 rad s/m

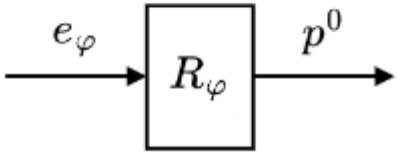
Y_δ= 0 m/s^2

With these assumptions in mind the lateral dynamics becomes:

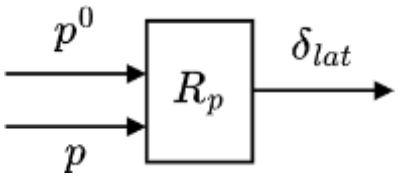


Task 1 – Attitude Control

- The two controllers consist of a proportional controller (R_φ) and a 2 DoF PID controller (R_p)
- They are modelled using the MATLAB functions:
  - TUNABLEPID (R_φ)

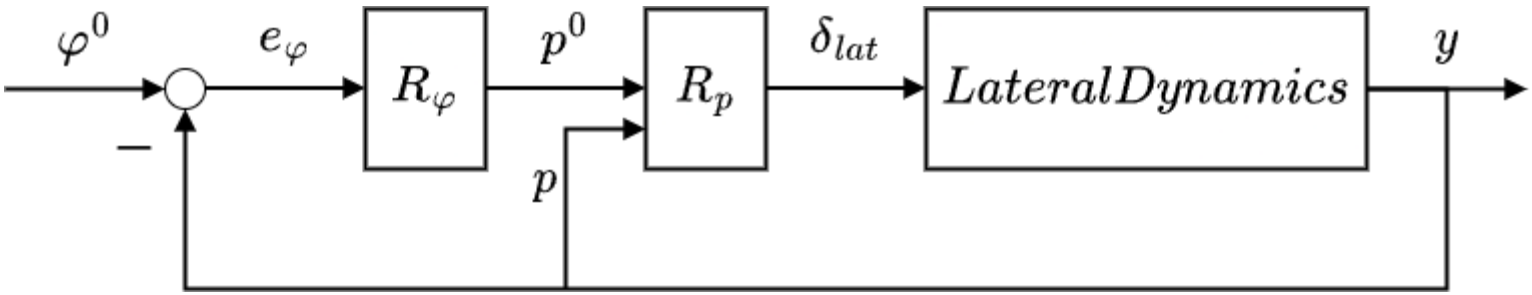


  - TUNABLEPID2 (R_p)



Task 1 – Attitude Control

- Calling «Lateral Dynamics» the aforementioned state space model we can draw the following block scheme:



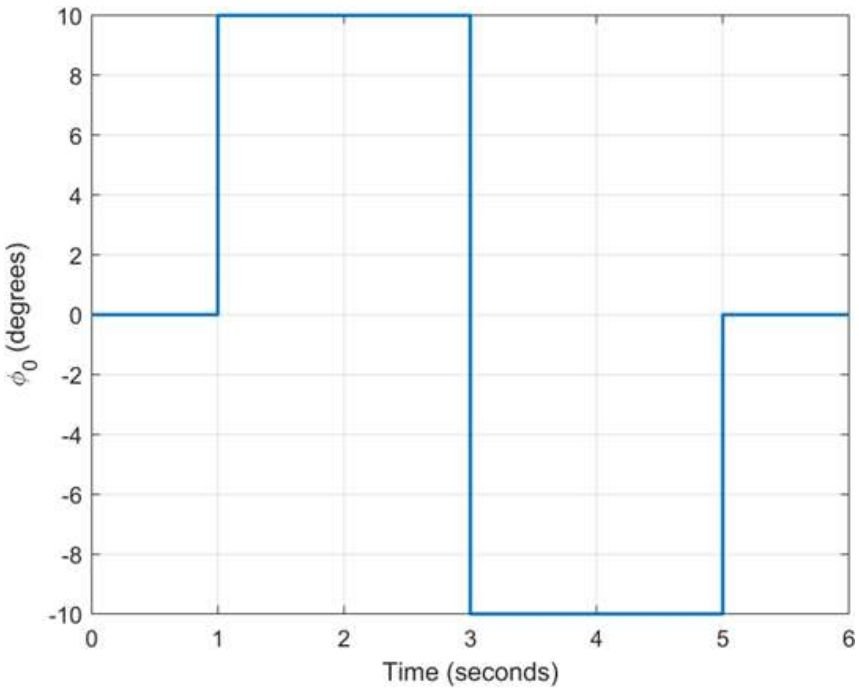
- The closed loop is created using the MATLAB function “connect” and its feature to make name-based interconnections.
- The p-component of the output vector y is fed back into the controller R_p

Task 1 – Attitude Control

- Performance Requirement

Response of φ to the variation of the setpoint must be equivalent to a second order system with

$\omega_n \geq 10\ \frac{rad}{s}$ and $\xi \geq 0.9$



- Control Effort Limitation

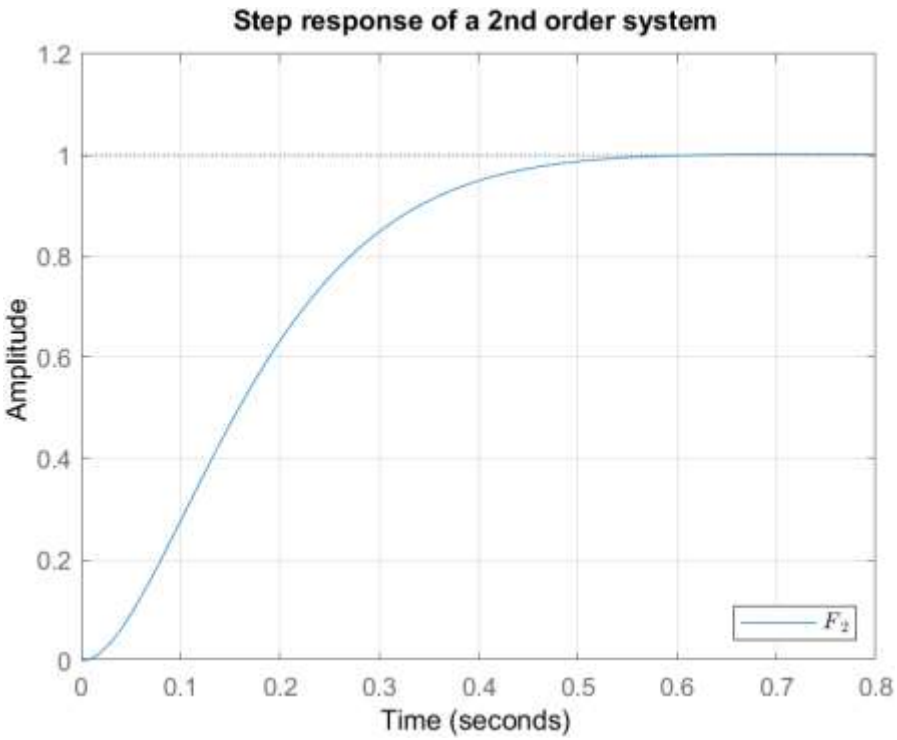
$|\delta| \leq 5\%$ for a doublet change of the setpoint φ of $\pm\ 10°$

Task 1 – Attitude Control

- A general second order system is expressed as:

$$F(s) = \frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}$$

- The step response of the reference system is the following:



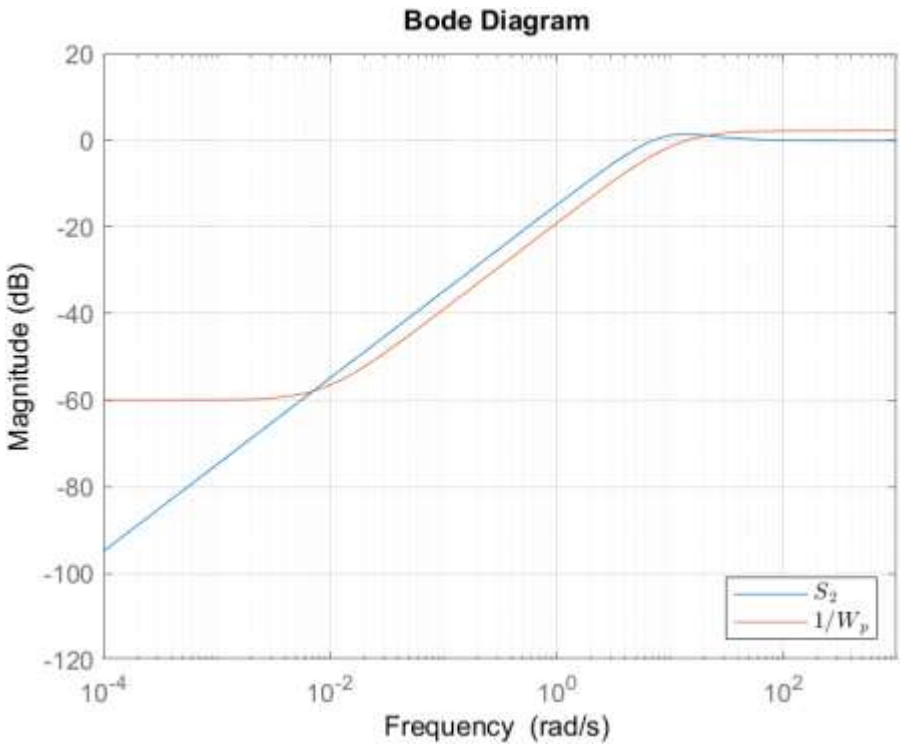
Task 1 – Attitude Control

- Performance Weight

$$W_p(s) = \frac{\frac{s}{M}+\omega_b}{s+A\omega_b}$$

with:

$$M = 1.3$$
$$A = 10^{-3}$$
$$\omega_b = 0.9 \times \omega_n$$



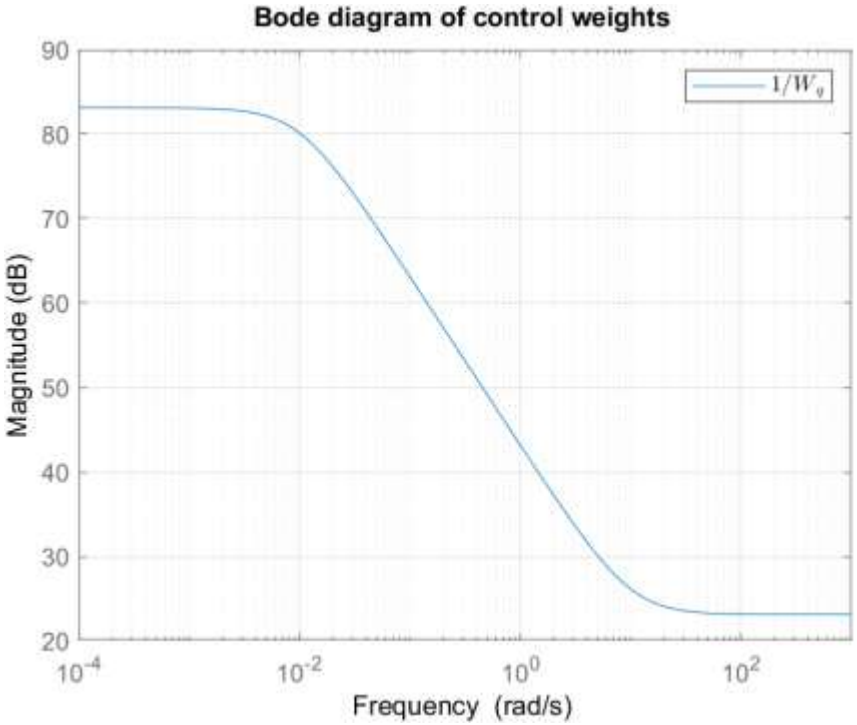
Task 1 – Attitude Control

- Control Effort Weight

$$W_q = \alpha \cdot \frac{s+\omega_\tau \cdot 10^{-3}}{s+\omega_\tau}$$

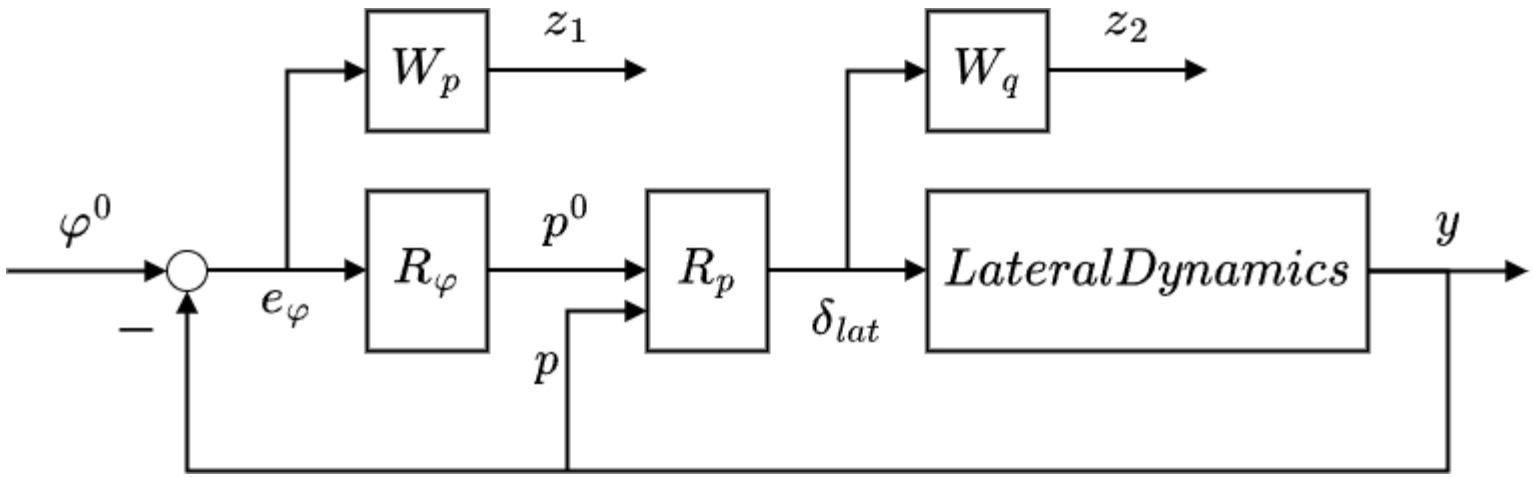
with:

$$\varphi_{max} = 20°$$
$$\mathrm{A} = deg2rad(\varphi_{max})$$
$$\alpha = A/5$$
$$\omega_\tau = 10\ \frac{rad}{s}$$



Task 1 – Attitude Control

- Including weighting functions, the block diagram becomes:



Task 1 – Attitude Control

- In MATLAB the "*hinfstruct*" function was used to tune the free parameters of the controllers. The tuning process performs a mixed sensitivity syntehsis minimizing the H_∞ norm of the closed-loop transfer function. The weighting functions that were previously designed are used to enforce performance and control requirements.

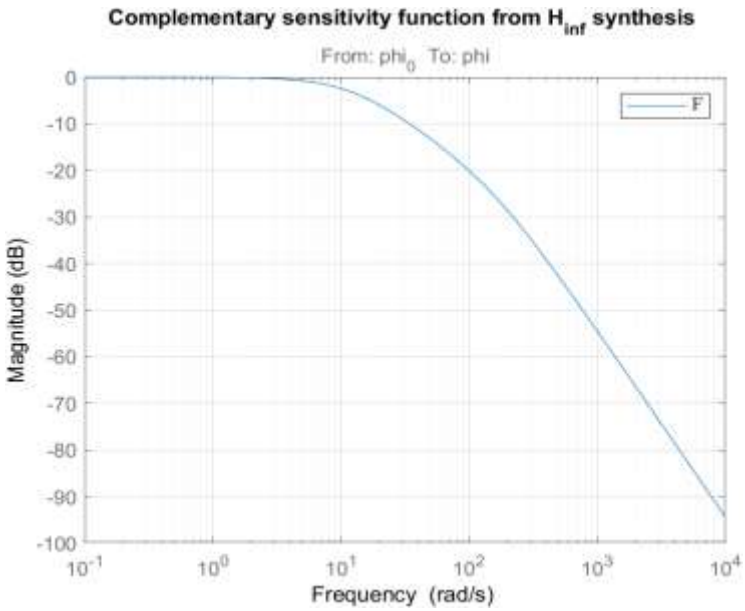
- Tuning Results:

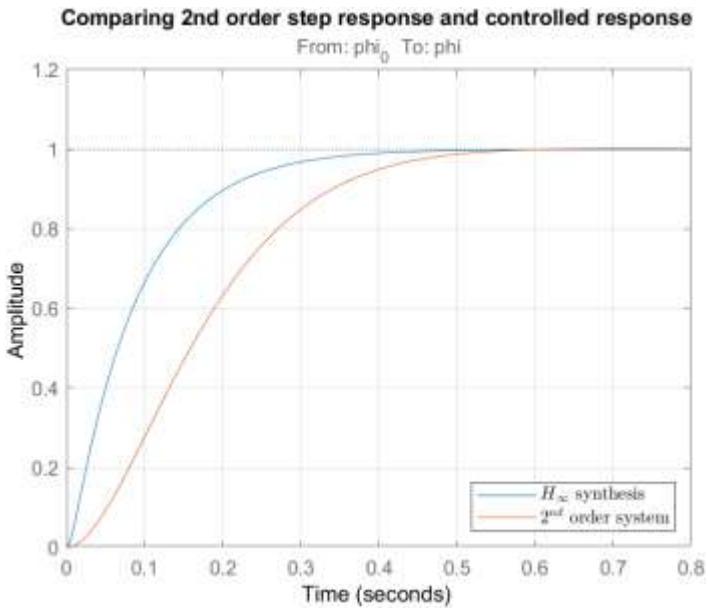
  - R_p:
    $K_p = 0.3876$
    $K_i = 1.0355$
    $K_d = -9.3897 \cdot 10^{-8}$

  - R_φ
    $K_p = 10.9213$

  - $\gamma = 0.824$

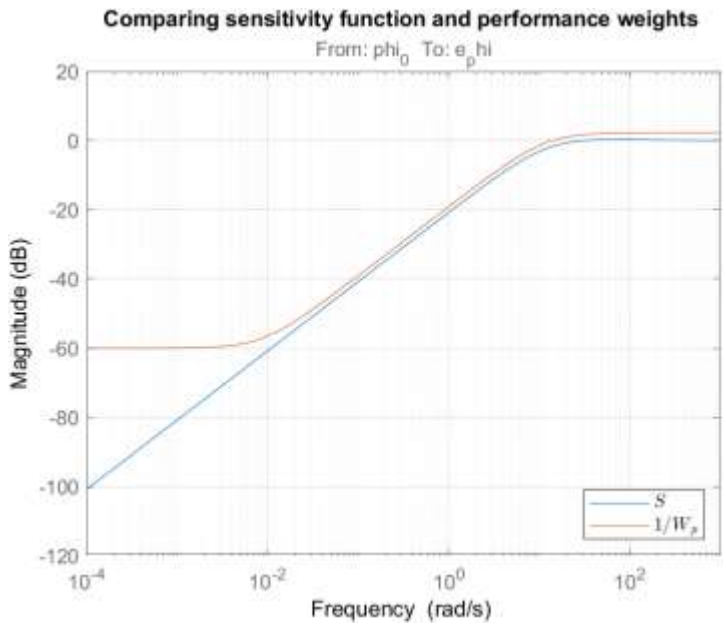
Task 1 – Attitude Control

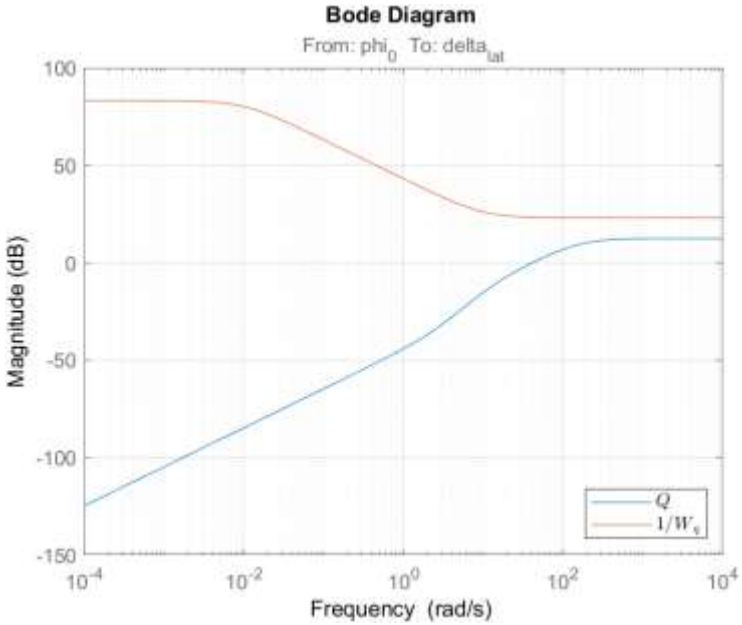




- Tuned closed loop step response has a better settling time, no overshoot and overall faster dynamics when compared to 2nd order approximation

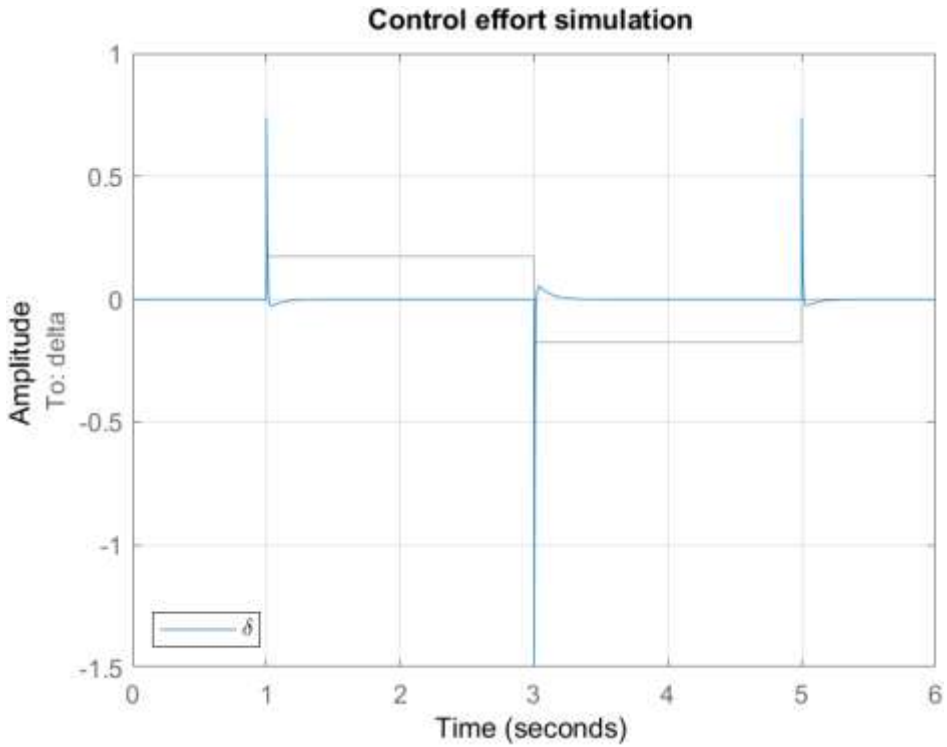
Task 1 - Attitude Control





- Both Sensitivity Function S & Control Sensitivity Function Q satisfy nominal performance requirements

Task 1 – Attitude Control



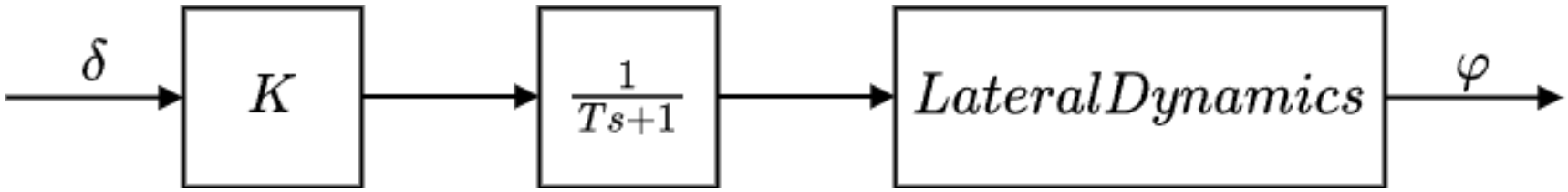
- Also through a time marching simulation we can see the control effort requirement is fulfilled

- Task 2 – Robustness Analysis

Task 2 – Robustness Analysis

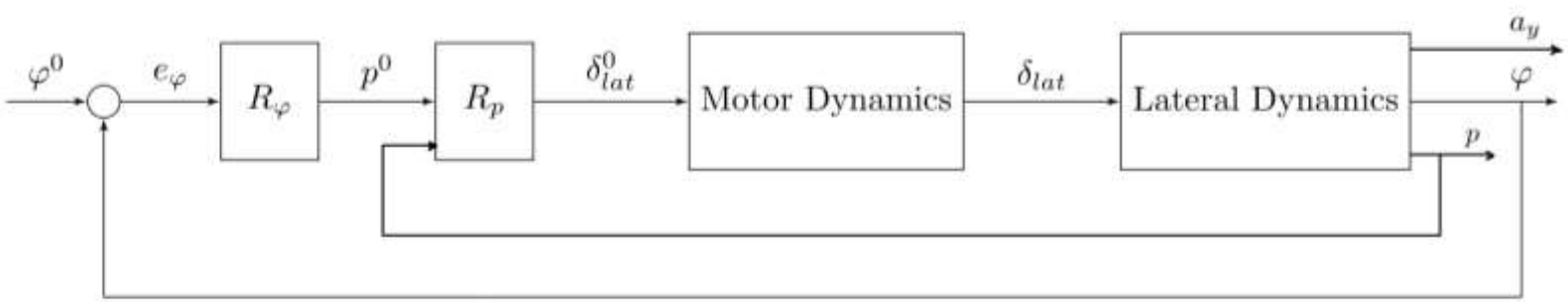
- From now on the dynamics of the motors is introduced by a transfer function with an assigned time constant $T = 0.02\ s$

- A gain $K = 1$ that can decrease down to 0.75 due to drop of battery voltage is added



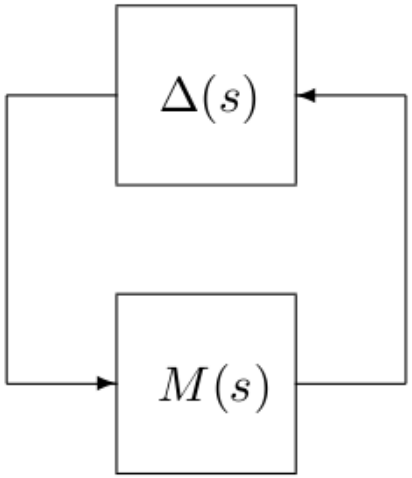
Task 2 – Robustness Analysis

- In order to perform robustness analysis taking into account the different uncertainties, our block scheme has to be written in the so called $M - \Delta$ form
- M is a block that contains all the nominal quantities of our system. Δ is a block that contains all the uncertainties related to the different parameters and that has to satisfy the condition $\Delta(j\omega) \leq 1 \; \forall \omega$



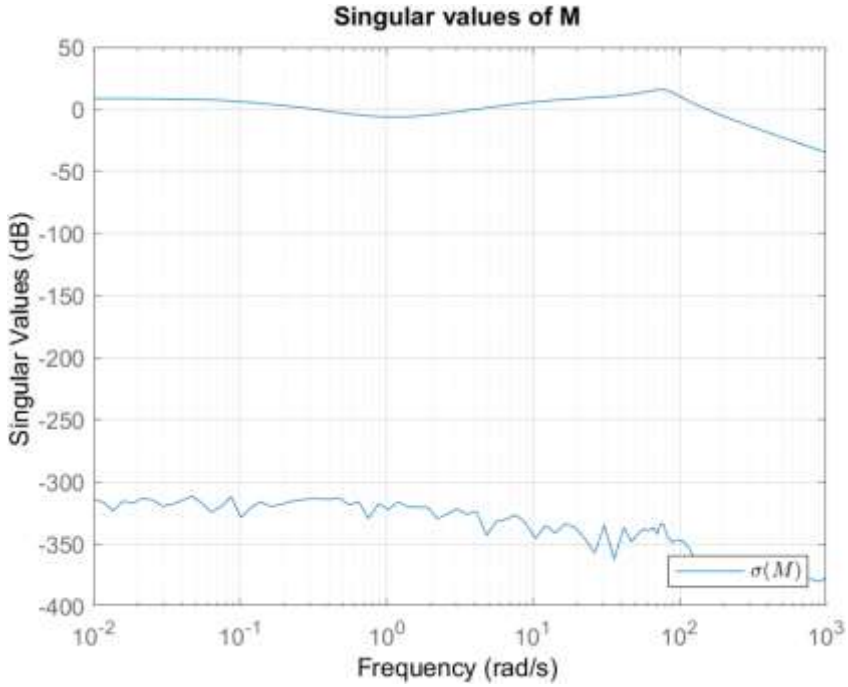
Task 2 – Robustness Analysis

- First we modelled the uncertainty through $W(s)$ and $\Delta(s)$ blocks, using the «*ucover*» MATLAB function
- To retrieve the the $M - \Delta$ form the MATLAB command «*lftdata*» is used to separate our uncertain system into a certain object M and a normalized uncertain matrix Δ



Task 2 – Robustness Analysis

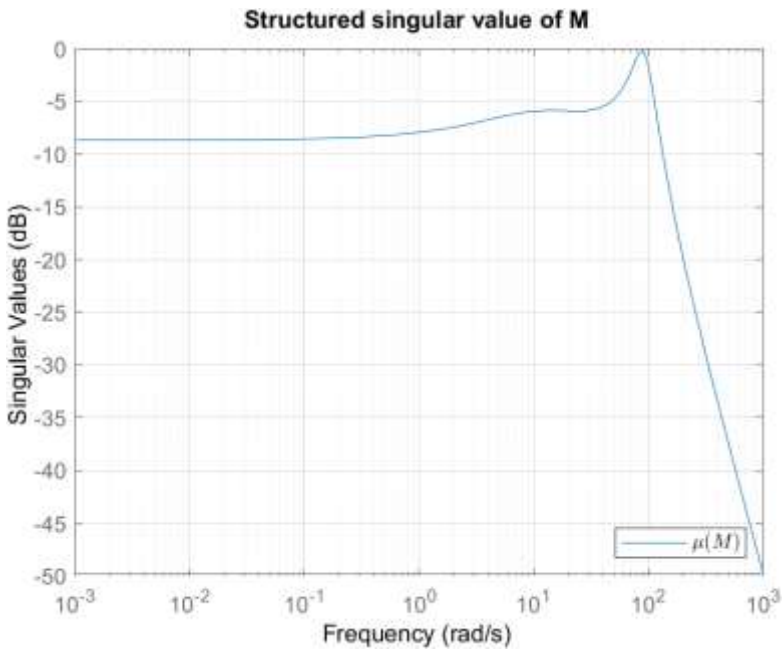
- Robust stability has first been checked computing the singular values of matrix M



- Robust stability condition $\sigma_{max}(M(j\omega)) < 1$ is not satisfied

Task 1 – Robustness Analysis

- Using MATLAB command «*mussv*» is possible to check robustness stability in terms of structured singular values of matrix M



- Now robust stability condition $\mu_D(M(j\omega)) < 1$ is satisfied

Task 2 – Robustness Analysis

- The last remaining step is the robust performance analysis. In a more general case RP can be proven by checking

7.6.2 Robust performance

For robust performance we require the performance condition (7.55) to be satisfied for *all* possible plants, that is, including the worst-case uncertainty.

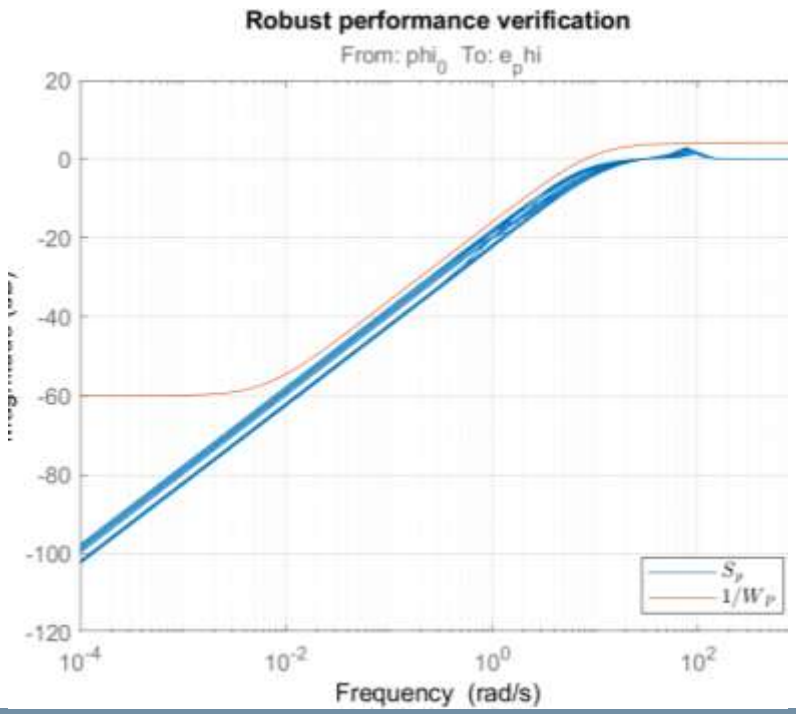
$$\text{RP} \overset{\text{def}}{\Leftrightarrow} \quad |w_P S_p| < 1 \quad \forall S_p, \forall \omega \tag{7.56}$$

$$\Leftrightarrow \quad |w_P| < |1 + L_p| \quad \forall L_p, \forall \omega \tag{7.57}$$

From Skogestad, «Multivariable Feedback and Control»

Task 2 – Robustness Analysis

- So we performed the robust performance analysis by comparing the Bode diagram of $^{1}/_{W_P}$ and of S_p (where capital P is «perfromance», whereas p is «perturbed»)



- Task 3 – Dynamics of the Observer

Task 3 – Dynamics of the Observer

- In reality the state φ is not measurable, so it is also required to add an observer in order to find a sufficiently accurate estimation of such quantity

- First of all Observability of the system has to be checked

```
% Check system observability
observability_matrix = obsv(A_nom, C_nom);
rank_observability = rank(observability_matrix);
if rank_observability < size(A_nom, 1)
    error('System is not observable.');
end
```

- Observability Matrix is of rank $n = 3$, so it is verified

Task 3: Dynamics of Observer

- First of all observer matrices have to be designed

```
L = place(A_nom', C_nom', [-10 -10 -10])';
```

$$A_{observer} = \begin{bmatrix} [A_{nom}] & [0] \\ [L \cdot C_{nom}] & [A_{nom} - L \cdot C_{nom}] \end{bmatrix}; \quad B_{observer} = \begin{bmatrix} [B_{nom}] \\ [B_{nom}] \end{bmatrix}$$

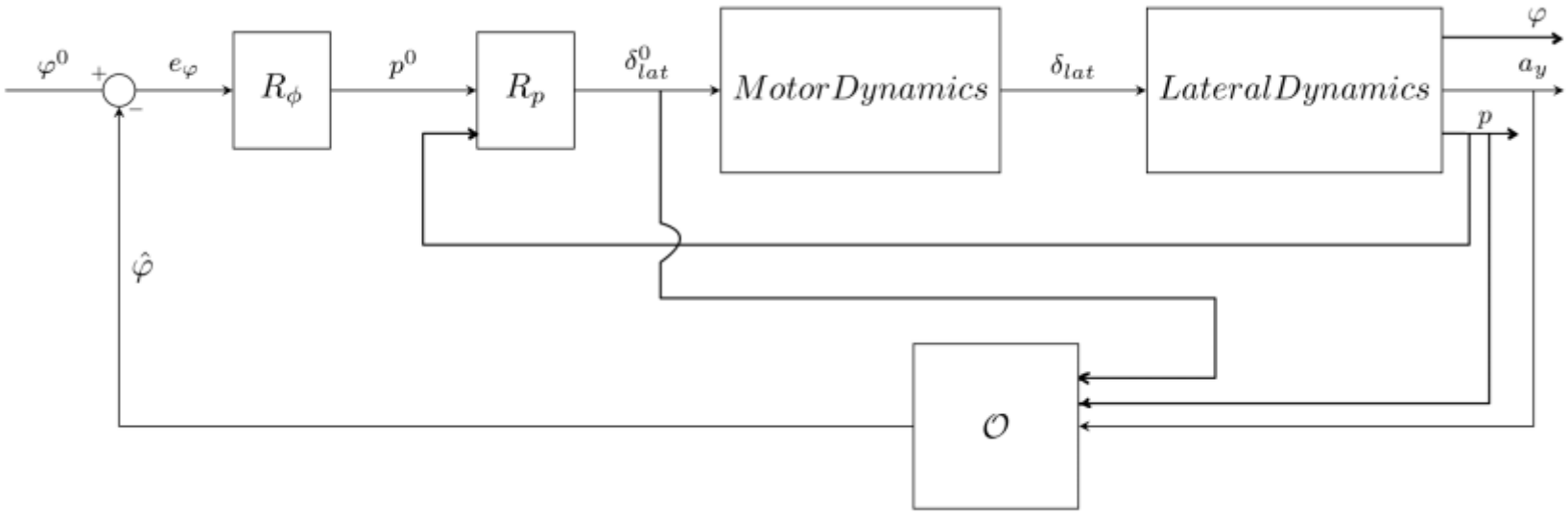
$$C_{observer} = \begin{bmatrix} [C_{nom}] & [0] \\ [0] & [C_{nom}] \end{bmatrix}; \quad D_{observer} = \begin{bmatrix} [D_{nom}] \\ [D_{nom}] \end{bmatrix}$$

- Then, a state-space form system can be created

```
observer_dyn = ss(A_observer, B_observer, C_observer, D_observer);
```

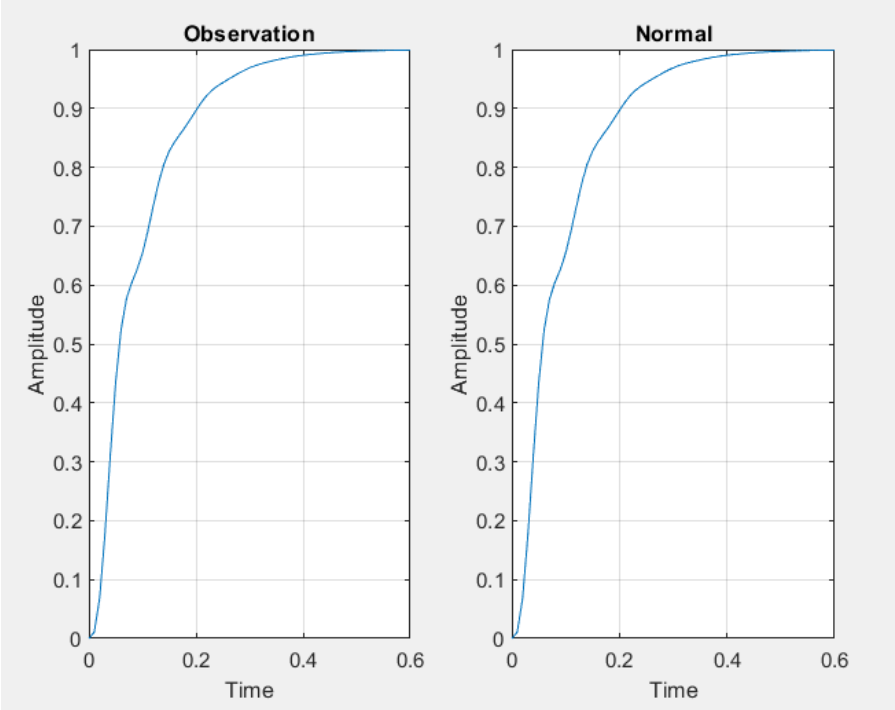
Task 3 – Dynamics of the Observer

- Using the MATLAB function «*connect*» the following block scheme with the observer was constructed



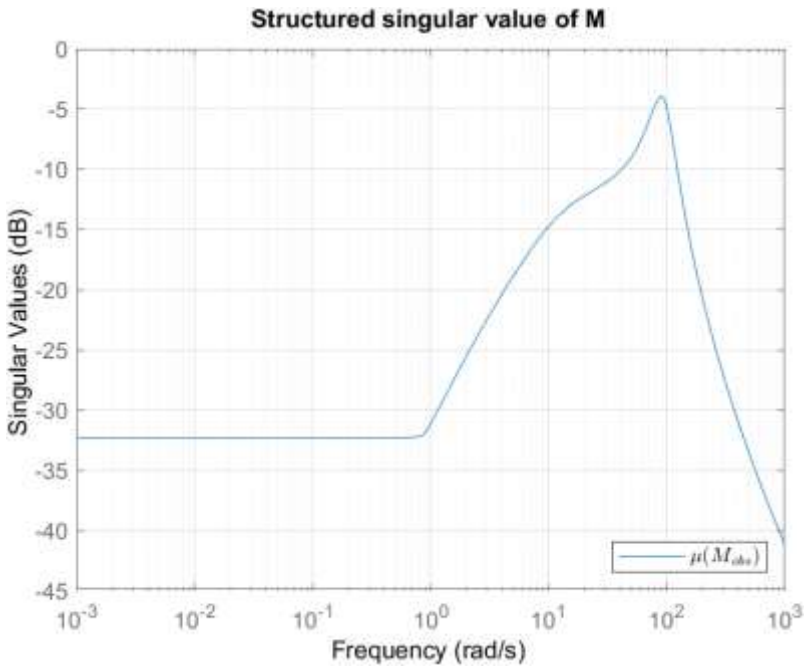
Task 3 – Dynamics of the Observer

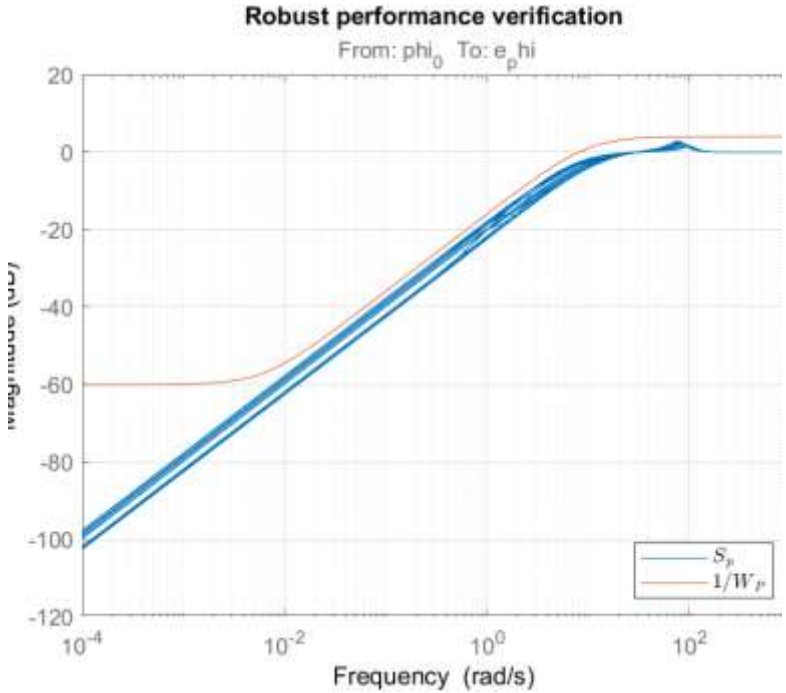
- To verify that the observer is correctly working, the step response was plotted and compared for both plant and plant with observer



Task 3 – Dynamics of the Observer

- In conclusion, we performed once again the robust stability and performace analysis as did before, this time considering the observer





- Task 4 – Monte Carlo Analysis

Task 4 – Monte Carlo Analysis

- Monte Carlo Analysis is a technique that simulates the behaviour of the system for a large set of possible combinations of uncertain parameters in order to gather statistical data. Such data can be then used to assess the fulfilment of requirements

- For the analysis the model has been considered without any control oriented assumptions, but also including motor dynamics, observer and battery gain uncertainty

Task 4 – Monte Carlo Analysis

- In practice this meant:

a) creating a for loop to be repeated for a large number of iterations

b) picking random values of the uncertain parameters (form the known information of them having Gaussian distribution)

c) simulating the response of such a system, composed of a random combination of uncertain parameters

d) collecting performance data in arrays and computing key statistics of such data

Task 4 – Monte Carlo Analysis

- The obtained results of the Monte Carlo Analysis have been collected into mean and standard deviation of three basic parameters, useful to express performance and control effort:

1. % overshoot for a step input

2. settling time for a step input

3. $|\delta|_{max}$ in response to a doublet setpoint

Task 4 – Monte Carlo Analysis

Monte Carlo Performance

- % overshoot

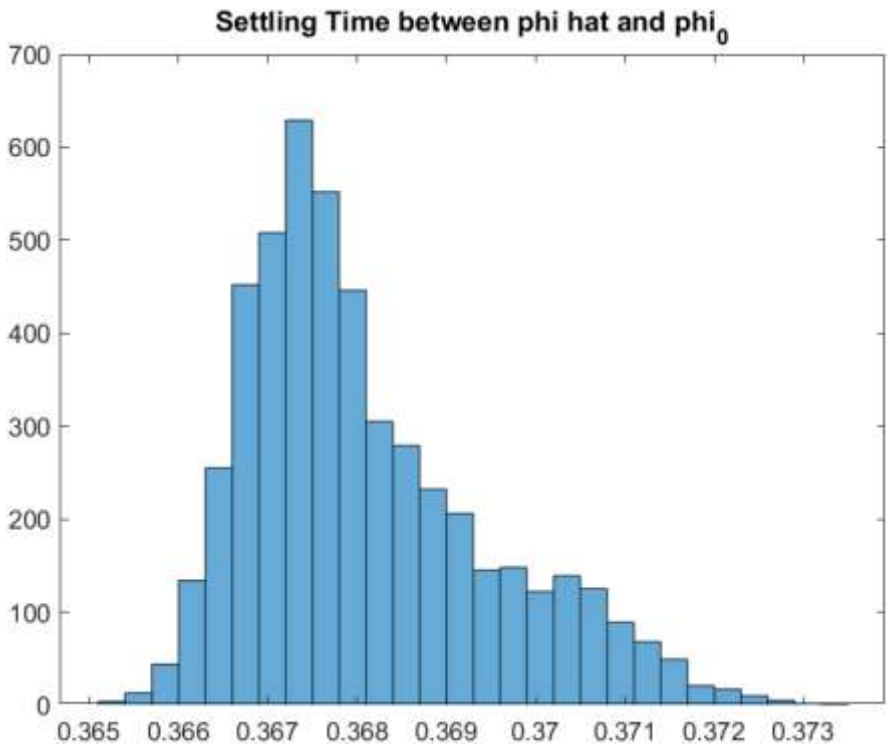
  $\mu_{\%} = 0\%$

  $\sigma_{\%} = 0\%$

- settling time

  $\mu_{\Delta t} = 0.3681\ s$

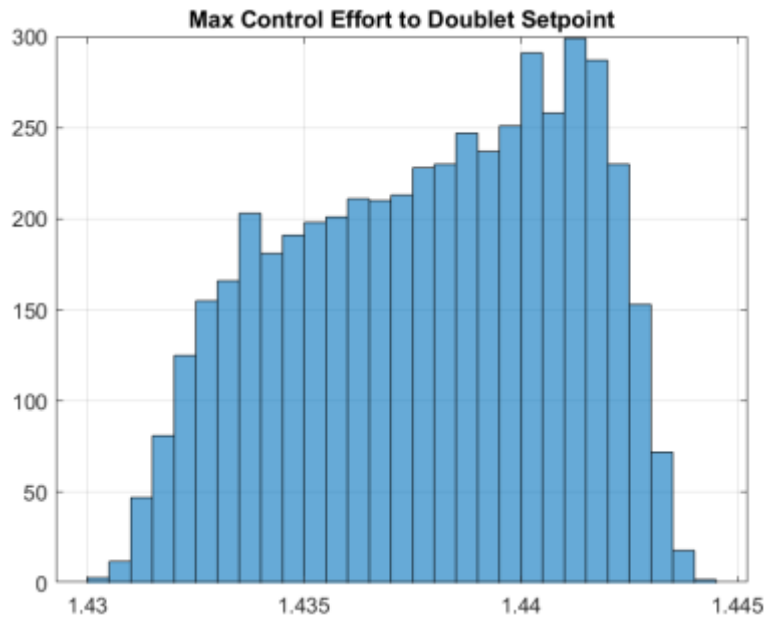
  $\sigma_{\Delta t} = 0.0014\ s$



Task 4 – Monte Carlo Analysis

Monte Carlo Control Effort

- $|\delta|_{max}$ in response to a doublet setpoint
  $\mu_\delta = 1.4379\ \%$
  $\sigma_\delta = 0.0032\ \%$



The end

Thank you for your attention,

Andreozzi Michele

Rinaldi Luca